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Controllability of the continuity equation for normalizing flows

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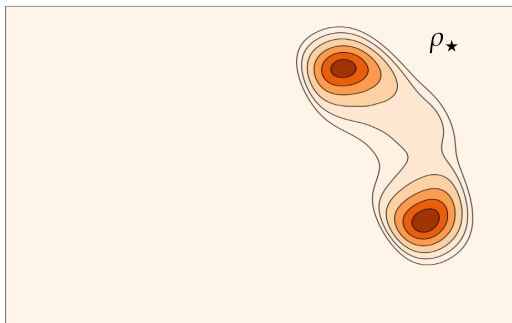
CHAPTER 1



Motivation

A problem from unsupervised learning

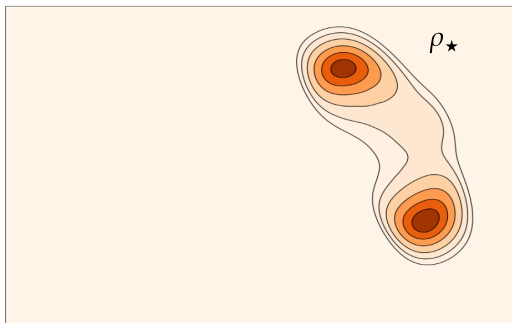
What is **flow-based** generative modeling?



A problem from unsupervised learning

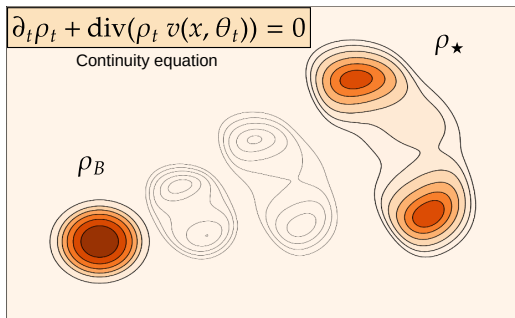
What is **flow-based** generative modeling?

Generating samples from a target distribution $\mu_\star$ (complex and high-dimensional) by transforming samples from a simple, base distribution μ_B via some learned map $\mathcal{T}$.



Continuous Normalizing Flows (CNFs)

- ◆ Base $\mu_B \equiv \mathcal{N}(0, \text{Id}) \mapsto$ Target $\mu_\star$ accessed through samples $x_1, \dots, x_N \sim \mu_\star$.
- ◆ Fix a Lipschitz control family $\{v_\theta(\cdot)\}_\theta$. Find $\theta(\cdot)$ such that $\mu_T^\theta = (\Phi_T^\theta)_\# \mu_B \approx \mu_\star$.
- ◆ A sample $x \sim \mu_B$ then follows the flow defined by
$$\begin{cases} \frac{d}{dt} \Phi_t^\theta(x) &= v_{\theta(t)}(\Phi_t^\theta(x)) \\ \Phi_0^\theta(x) &= x \end{cases}$$



Continuous Normalizing Flows (CNFs)

Many SOTA models share this continuity-equation formulation, differing in their training objectives.

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a) Likelihood-trained.¹ Maximize the likelihood at the sample, $\prod_i \rho_t^\theta(x_i)$, computed by solving

$$\frac{d}{dt} \log \rho_t^\theta(\Phi_t^\theta(x)) = -\operatorname{div} v_{\theta(t)}(\Phi_t^\theta(x)).$$

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b) Flow matching.² Prescribe particle paths X_t which induce a field u_t , then apply regression:

$$\min_{\theta} \mathbb{E}_{t, X_t \sim \rho_t} \left[\|v_{\theta(t)}(X_t) - u_t(X_t)\|^2 \right].$$

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c) Score-based diffusion models.³ A forward SDE transforms data into noise. Learn the score $\nabla \log \rho_t$ to run the reverse-time SDE $\implies$ Density solves a Fokker-Planck equation, but the same marginals are generated by the **probability flow ODE** (hence, by a continuity eq.).

¹ R. T. Q. Chen et al. NeurIPS (2018); W. Grathwohl et al. ICLR (2019).

² Y. Lipman et al. ICLR (2023); X. Liu et al. ICLR (2023); M. S. Albergo & E. Vanden-Eijnden, ICLR (2023).

³ J. Sohl-Dickstein et al. ICML (2015); J. Ho et al. NeurIPS (2020); Y. Song et al. ICLR (2021).

CHAPTER 2



Problem

Problem: KL controllability

The **Kullback-Leibler (KL) divergence** between two probability measures μ, ν is

$$\text{KL}(\mu \parallel \nu) := \begin{cases} \int_{\mathbb{R}^d} \log \left(\frac{d\mu}{d\nu} \right) d\mu = \int_{\mathbb{R}^d} \log \left(\frac{\rho_\mu}{\rho_\nu} \right) \rho_\mu dx, & \mu \ll \nu, \\ +\infty, & \text{otherwise.} \end{cases}$$

The **optimization problem** for a CNF can be equivalently written, at the population level, as

$$\min_{\theta} \text{KL}(\mu_{\star} \parallel \mu_T^{\theta})$$

Why? $\max_{\theta} \mathcal{L}_N(\theta) = \max_{\theta} \prod_{i=1}^N \rho_T^{\theta}(x_i) \iff \max_{\theta} \frac{1}{N} \sum_{i=1}^N \log \rho_T^{\theta}(x_i)$

$$\min_{\theta} \text{KL}(\mu_{\star} \parallel \mu_T^{\theta}) = \min_{\theta} \left(-\mathbb{E}_{\mu_{\star}} [\log \rho_T^{\theta}] + \underbrace{\mathbb{E}_{\mu_{\star}} [\log \rho_{\star}]}_{\text{indep. of } \theta} \right) = \max_{\theta} \mathbb{E}_{\mu_{\star}} [\log \rho_T^{\theta}]$$

$\downarrow \text{LLN}$

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We can view CNF training as an **approximate controllability problem**:

Given $\varepsilon > 0$ and $T > 0$, find a control $\theta = (\theta(t))_{t \in [0, T]}$ such that

$$\text{KL}(\mu_\star \parallel \mu_T^\theta) < \varepsilon, \quad \text{subject to} \quad \begin{cases} \partial_t \rho_t^\theta(x) + \text{div}(\rho_t^\theta(x) v_{\theta(t)}(x)) = 0, \\ \rho_0(x) = \rho_B(x), \end{cases}$$

where the vector field $v_{\theta(t)}$ is a perceptron:

$$v_{\theta(t)}(x) = w(t)(a(t)^\top x + b(t))_+, \quad (w(t), a(t), b(t)) \in \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}.$$

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- ◆ **Difficulty 1: KL is not a distance.** It is non-symmetric and has no triangle inequality.
- ◆ **Difficulty 2: KL may blow up.** We need

$$\mu_\star \ll \mu_T^\theta \quad \text{and sufficient control of} \quad \frac{\rho_\star}{\rho_T^\theta} \quad \text{in the tails.}$$

$\Rightarrow$ *We need control over the support and the relative tail decay.*

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What about empirical targets? Given samples $x_1, \dots, x_N \sim \mu_\star$, the empirical measure is discrete, whereas our analysis considers an **absolutely continuous target**. A natural approach is to regularize:

$$\hat{\rho}_{\star, N}(x) = \frac{1}{N} \sum_{i=1}^N K_h(x - x_i), \quad \hat{\mu}_{\star, N} \approx \mu_\star.$$

Related work: transport by controlled flows

- Moser, Trans. AMS (1965);
Dacorogna--Moser, AIHP (1990).
Density-to-density diffeomorphisms and prescribed-Jacobian constructions.
- Khesin-Lee, J. Symplectic Geom. (2009).
Nonholonomic Moser theorem: transport along bracket-generating distributions.
- Agrachev-Caponigro, AIHP/JDCS (2009--10).
Controllability on the group of diffeomorphisms via time-varying feedback.
- Brockett, Lect. Notes Math. (2012).
Ensemble control: open-loop vs. feedback; moment controllability for linear systems.
- Duprez-Morancey-Rossi, SICON (2019).
Approximate controllability with localized Lipschitz vector fields; exact control at lower regularity.
- Duprez-Morancey-Rossi, JDE (2020).
Minimal time for controllability by localized perturbations of the velocity.
- Ruiz-Balet-Zuazua, SIAM Rev. (2023).
Constructive controllability of neural ODEs in the Wasserstein metric.
- Ishikawa et al., JMLR (2023).
Universality of flows via approximation of diffeomorphisms.
- Elamvazhuthi et al., J. Comput. Dyn. (2023).
Dynamical optimal transport under nonlinear control-affine constraints.
- Ruiz-Balet-Zuazua, JMPA (2024).
Constructive controllability of neural ODEs in TV metric.
- Raginsky, Geom. & Topol. Control Syst. Design (2025).
Controllability through feedback realization of diffeomorphisms.
- Scagliotti-Farinelli, Nonlinear Anal. (2025).
Approximation of W_2 -optimal maps by linear-control ODE flows.
- Caponigro-Vicari, JDCS (2025).
Controllability of the continuity equation from simultaneous point controllability.

CHAPTER 3



Main result

Main result

Theorem (A. Á-L, B. Geshkovski, D. Ruiz-Balet)

Let $\mu_B = \mathcal{N}(m_B, \Sigma_B)$, and let $\rho_\star$ be a probability density with $\rho_\star \log \rho_\star \in L^1(\mathbb{R}^d)$ and

$$\exists M, \sigma_\bullet > 0 : \quad \rho_\star(x) \leq \rho_\bullet(x) := \frac{1}{(2\pi\sigma_\bullet^2)^{d/2}} e^{-\|x\|^2/(2\sigma_\bullet^2)} \quad (\|x\| \geq M).$$

Then for all $T > 0$ and $\varepsilon > 0$ there exists a piecewise-constant control θ such that

$$\text{KL}(\mu_\star \parallel \mu_T^\theta) = \int_{\mathbb{R}^d} \rho_\star \log \frac{\rho_\star}{\rho_T^\theta} < \varepsilon.$$

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Number of switches: one can choose θ with L discontinuities, where

$$L(\varepsilon) \leq (d+10) \underbrace{\left\lceil \frac{2R_\varepsilon}{h_\varepsilon} \right\rceil^d}_{\text{number of cells}} + 2d$$

Here $[-R_\varepsilon, R_\varepsilon]^d$ contains at least $1 - \varepsilon$ mass of both μ_B and $\mu_\star$, and h_ε is the mesh size of a uniform grid for which the Riemann approximation has TV-error at most ε .

Proof strategy

(i) **Approximate¹ in L^1 /TV on $[0, T/2]$.** Construct $\theta(\cdot)$ such that

$$|\mu_{T/2}^\theta - \mu_\star|_{\text{TV}} = \frac{1}{2} \|\rho_{T/2}^\theta - \rho_\star\|_{L^1} < \varepsilon_0 \quad \text{and} \quad \rho_{T/2}^\theta > 0.$$

(ii) **Repair the tails on $[T/2, T]$.** Leave a large cube unchanged and achieve

$$\rho_T^\theta(x) \geq \rho_\star(x) \quad \text{for all } \|x\| \gg 1.$$

(iii) **Conclude with reverse Pinsker inequality².** The L^1 error remains small by (i), and $\|\mathrm{d}\mu_\star/\mathrm{d}\mu_T^\theta\|_{L^\infty} =: C_\theta < \infty$ by (ii). Therefore we can apply

$$\text{KL}(\mu_\star \|\mu_T^\theta) \leq \frac{\log C_\theta}{1 - C_\theta^{-1}} \cdot |\mu_\star - \mu_T^\theta|_{\text{TV}} = \frac{1}{2} \cdot \frac{\log C_\theta}{1 - C_\theta^{-1}} \cdot \|\rho_\star - \rho_T^\theta\|_{L^1}.$$

¹ D. Ruiz-Balet and E. Zuazua. J. Math. Pures Appl. (2024).

² I. Sason. On Reverse Pinsker Inequalities (2015).

Tool I: Closed-form solution

The solution of the continuity equation

$$\partial_t \rho_t^\theta(x) + \operatorname{div} \left(\rho_t^\theta(x) w (a^\top x + b)_+ \right) = 0, \quad \rho_0 = \rho_B$$

has a **closed-form** (by the method of characteristics):

$$\rho_t^\theta(x) = \begin{cases} \rho_B(x), & x \in H^- := \{a^\top x + b \leq 0\}, \\ e^{-t w^\top a} \rho_B(A_t(x)), & x \in H^+ := \{a^\top x + b > 0\}, \end{cases}$$

where the transformation $A_t(x)$ is **affine** in x :

$$(w^\top a \neq 0) : \quad A_t(x) = e^{-t w a^\top} x - \frac{b}{w^\top a} (1 - e^{-t w^\top a}) w, \quad (1)$$

$$(w^\top a = 0) : \quad A_t(x) = (I - t w a^\top) x - t b w. \quad (2)$$

A ReLU flow leaves one half-space fixed and applies an affine deformation to the other.

Tool II: L^1 -stability

(**Lemma**). Let $\eta \in L^1(\mathbb{R}^d)$ satisfy

$$\|\eta\|_{L^1(\mathbb{R}^d)} < \varepsilon,$$

and consider the initial data

$$\rho_0^\eta = \rho_B + \eta, \quad \rho_0^0 = \rho_B.$$

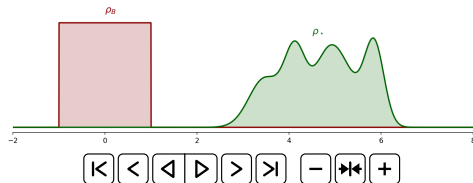
Fix $T > 0$, let $v : [0, T] \rightarrow \text{Lip}(\mathbb{R}^d; \mathbb{R}^d)$ be piecewise constant in time and consider

$$\partial_t \rho_t + \text{div}(\rho_t v_t(\cdot)) = 0. \tag{3}$$

Let ρ_t^η and ρ_t^0 solve (3) with initial data ρ_0^η and ρ_0^0 , respectively. Then:

$$\|\rho_t^\eta - \rho_t^0\|_{L^1(\mathbb{R}^d)} = \|\eta\|_{L^1(\mathbb{R}^d)} < \varepsilon, \quad t \in [0, T].$$

Step 1: L^1 control in dimension $d = 1$

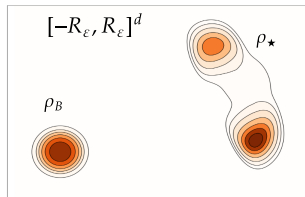


Basic mechanism:

- ◆ Approximate the target by K disjoint piecewise-constant blocks.
- ◆ Approximate a uniform reference density by K blocks with the same mass.
- ◆ ReLU flows generate localized **compression/dilations and translations**.
- ◆ Control the blocks inductively, one by one.

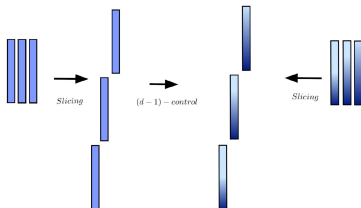
L^1 -stability transfers the exact control of the approximations to approximate control of the original densities.

Step 1: L^1 control in dimension $d > 1$



- Discretize.** Approximate both densities on the same Cartesian mesh and remove thin strips between neighboring cells.
- Use reversibility.** Reduce the problem to controlling a uniform density on the grid towards a piecewise-constant one.

Step 1: L^1 control in dimension $d > 1$



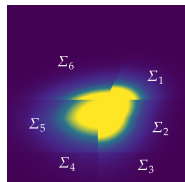
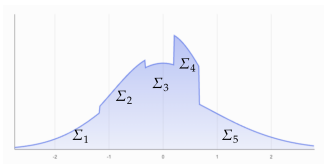
- a) **Discretize.** Approximate both densities on the same Cartesian mesh and remove thin strips between neighboring cells.
- b) **Use reversibility.** Reduce the problem to controlling a uniform density on the grid towards a piecewise-constant one.
- c) **Slice.** Separate the x_d -slices by translations parallel to $x_d = \text{const}$ so that their projections onto $\mathbb{R}^{d-1}$ are disjoint.
- d) **Reduce dimension.** Project onto the first $d - 1$ coordinates and apply c) in $\mathbb{R}^{d-1}$.

Iterating the argument ($d \rightarrow d - 1 \rightarrow \dots$) eventually reduces the problem back to $d = 1$.

Step 2: Tail repair

After Step 1, $\rho_{T/2}^\theta$ is a **piecewise Gaussian**. Choose a smaller Gaussian

$$\underline{\rho}(x) = \alpha e^{-\|x\|^2/(2\underline{\sigma}^2)} \quad \text{with} \quad \underline{\rho}(x) < \rho_{T/2}^\theta(x) \quad \text{for all } x.$$

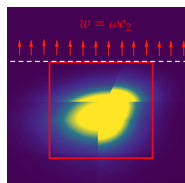
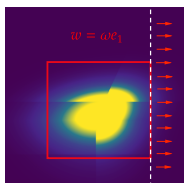


Evolution of a Gaussian in 1D and 2D after applying finitely many constant controls.

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Apply to $\underline{\rho}$ one control outside each of the $2d$ faces of a large cube:

- ◆ the density is **unchanged inside the cube**;
- ◆ outside, the flow **increases the variance** in the outward direction;
- ◆ after the $2d$ controls,

$$\underline{\rho}_T(x) \geq \rho_\bullet(x) \underset{\text{(by hyp.)}}{\geq} \rho_\star(x) \quad \text{for } \|x\| \gg 1.$$

Conclusion of the proof

Let $Q = [-\overline{M}, \overline{M}]^d$ be the cube left unchanged in Step 2.

The L^1 error remains small:

$$\|\rho_T^\theta - \rho_\star\|_{L^1} \leq \int_Q |\rho_{T/2}^\theta - \rho_\star| + \int_{Q^c} \rho_T^\theta + \int_{Q^c} \rho_\star < 3\varepsilon_0.$$

Moreover, the tail construction gives

$$\frac{\rho_\star(x)}{\rho_T^\theta(x)} \longrightarrow 0 \quad \text{as } \|x\| \rightarrow \infty,$$

while $\rho_T^\theta > 0$. Hence C_θ is uniformly bounded, and reverse Pinsker yields

$$\text{KL}(\mu_\star \parallel \mu_T^\theta) < \varepsilon.$$

*Arbitrarily accurate TV control + relative tail control + positivity
 $\implies$ KL controllability.*

CHAPTER 4



Extensions

Reverse KL (variational inference)

Theorem (A. Á-L, B. Geshkovski, D. Ruiz-Balet)

Assume $\rho_\star > 0$, $\rho_\star \log \rho_\star \in L^1$ and, for some $M, \sigma_\bullet > 0$,

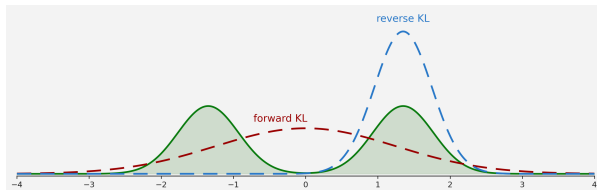
$$\rho_\star(x) \geq \rho_\bullet(x) = \frac{1}{(2\pi\sigma_\bullet^2)^{d/2}} e^{-\|x\|^2/(2\sigma_\bullet^2)} \quad (\|x\| \geq M).$$

Then, for all $T, \varepsilon > 0$, there exists a piecewise-constant control θ such that

$$\text{KL}(\mu_T^\theta \parallel \mu_\star) < \varepsilon.$$

Same construction, but now producing **TV-approximation** together with $\|\rho_T^\theta/\rho_\star\|_{L^\infty} < \infty$, so that

Reverse Pinsker:
$$\text{KL}(\mu_T^\theta \parallel \mu_\star) \leq \frac{1}{2} \frac{\log \|\rho_T^\theta/\rho_\star\|_{L^\infty}}{1 - \|\rho_T^\theta/\rho_\star\|_{L^\infty}^{-1}} \|\rho_\star - \rho_T^\theta\|_{L^1} < \varepsilon.$$



Beyond Gaussians

- ◆ Write the densities in Gibbs form:

$$\rho_B(x) = e^{-U_B(x)}, \quad \rho_*(x) = e^{-U_*(x)}.$$

- ◆ **Assumptions:**

- a) Linear growth:

$$U_*(x) \gtrsim \|x\| \quad \text{for } \|x\| \gg 1.$$

- b) **Compatibility:** given $A \in \mathbb{R}^{d \times d}$ with $\sigma(A) \subset (0, +\infty)$, there exists $\lambda_A > 0$ s.t.

$$\frac{\text{conv } U_*(\lambda_A x)}{U_B(Ax)} \rightarrow +\infty \quad \text{as } \|x\| \rightarrow \infty.$$

This compatibility condition enables the relative tail control needed after the affine transformations generated by the flow.

Our extension: If (a) and (b) hold, then for every $T > 0$ and $\varepsilon > 0$, there exists a piecewise-constant control θ with finitely many switches such that

$$\text{KL} \left(\mu_* \parallel \mu_T^\theta \right) < \varepsilon.$$

Beyond KL

For any convex $f : (0, \infty) \rightarrow \mathbb{R}$ with $f(1) = 0$, the **Csiszár f -divergence** is defined by

$$D_f(\mu_1, \mu_2) := \int f\left(\frac{d\mu_1}{d\mu_2}\right) d\mu_2.$$

◆ Some possible choices.

- ◆ $f(t) = t \ln t \Rightarrow$ KL divergence.
- ◆ $f(t) = \frac{1}{2}|t - 1| \Rightarrow$ Total variation.
- ◆ $f(t) = (\sqrt{t} - 1)^2 \Rightarrow$ Squared Hellinger distance.
- ◆ $f(t) = (t - 1)^2 \Rightarrow \chi^2$ -divergence.

◆ Our main result extends as:

$$\sup_{t>0} \frac{f(t)}{t \ln t} < \infty \quad \implies \quad D_f(\mu_\star \parallel \mu_T^\theta) < \varepsilon.$$

Beyond Lipschitz

Motivation

Can we match densities with **different tail regimes** such as $e^{-|x|^p} \rightarrow e^{-|x|^q}$?

It is possible by breaking global Lipschitz constraint of the vector field!

- ◆ Use the **locally Lipschitz** activation $\sigma(x) = x \log x \cdot \mathbf{1}_{\{x>1\}}$:

$$\partial_t \rho_t(x) + \partial_x (\sigma(x) \rho_t(x)) = 0, \quad \rho_0(x) \propto e^{-|x|^p},$$

- ◆ Characteristics increase exponentially but **don't blow up in finite time**:

$$\Phi_t^\theta(x) = \begin{cases} x, & x \leq 1, \\ x^{e^t}, & x > 1 \end{cases} \quad \rho_t^\theta(x) = \begin{cases} \rho_B(x), & x \leq 1, \\ e^{-t} \rho_B(x^{e^{-t}}) x^{e^{-t}-1}, & x > 1. \end{cases}$$

For $0 < q \leq p$, choosing $T \geq \log(p/q)$ yields

$$\lim_{x \rightarrow +\infty} \frac{\rho_T(x)}{e^{-x^q}} < +\infty.$$

Changing the sign of the vector field gives the analogous result for $q \geq p$.

Thank you!



Paper

Constructive approximate transport maps with normalizing flows

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Takeaways

- ◆ One-neuron ReLU perceptrons suffice for approximate transport in KL.
- ◆ The construction is explicit, with quantitative bounds on the number of switches.
- ◆ The framework extends to reverse KL, non-Gaussian bases and other divergences.
- ◆ Locally Lipschitz dynamics allow us to move between different tail regimes.